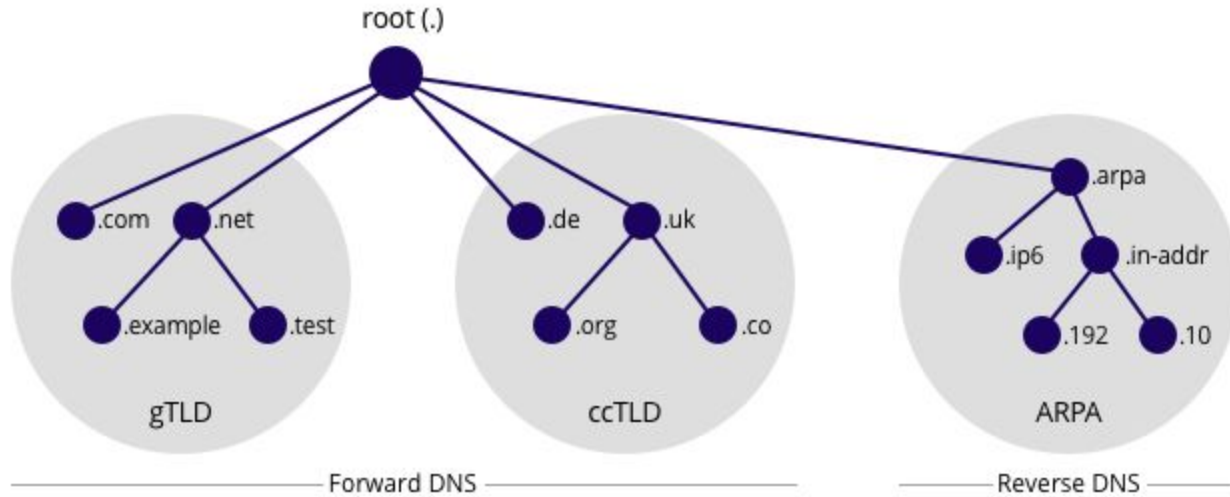


# IT Infrastructure services

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# DNS



# DNS record types

- A
- AAAA
- CNAME
- NS
- SOA
- MX
- TXT
- Many more: [https://en.wikipedia.org/wiki/List\\_of\\_DNS\\_record\\_types](https://en.wikipedia.org/wiki/List_of_DNS_record_types)

# DNS troubleshooting

- <https://isitdns.com/>
- nslookup <name>
- host <ip or name> [dns-server]
- dig [type] <name> [@dns-server]

# Bind9

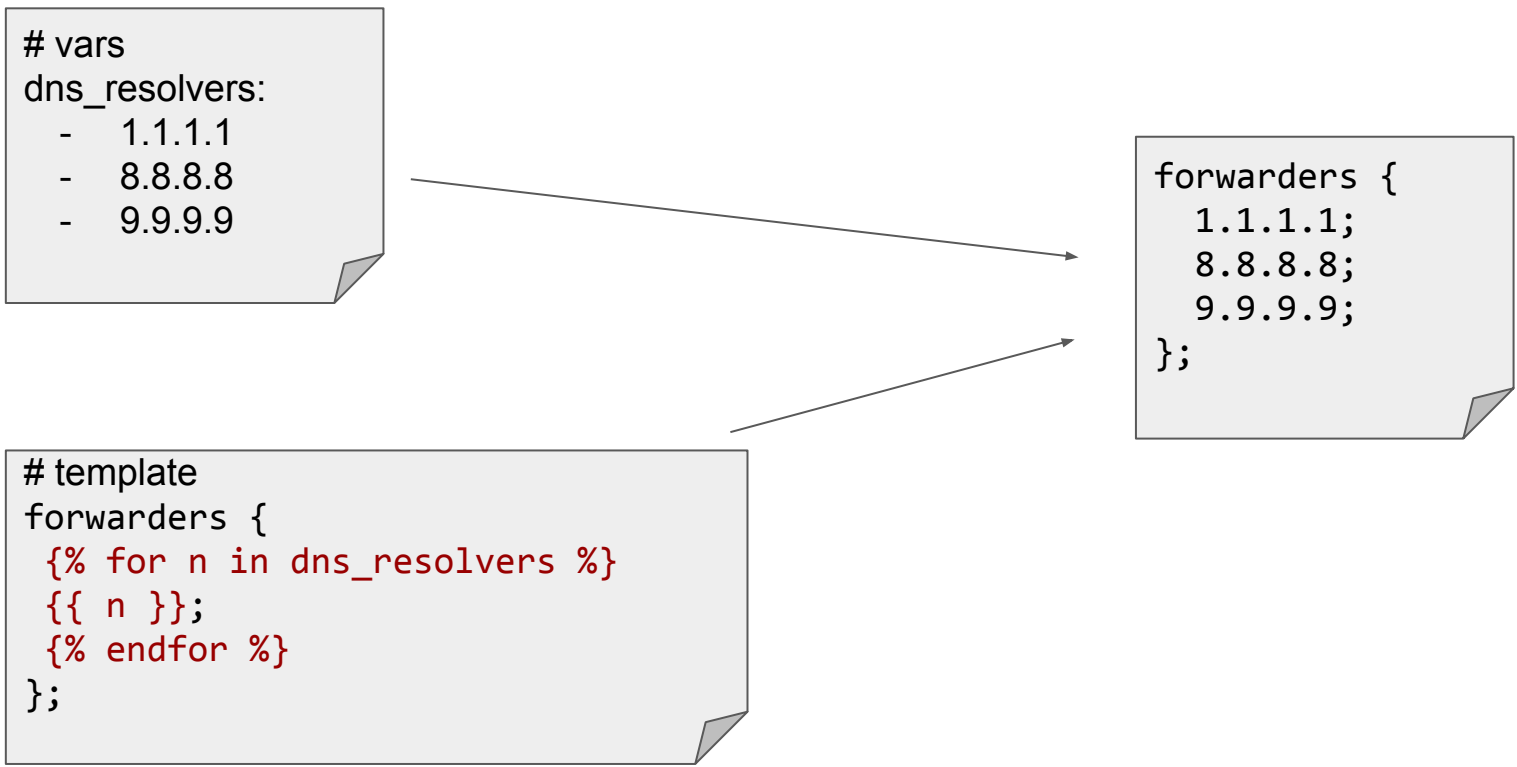
Docs: <https://kb.isc.org/docs/aa-01031>



# Jinja2 loops

Jinja2 online compiler: <https://j2live.ttl255.com/>

```
# vars
dns_resolvers:
  - 1.1.1.1
  - 8.8.8.8
  - 9.9.9.9
```

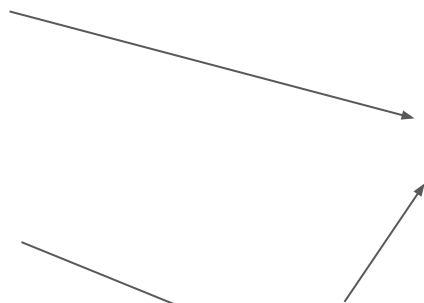
A diagram illustrating the Jinja2 loop rendering process. It consists of three code blocks. The top-left block shows a variable 'dns\_resolvers' with a list of three IP addresses. The bottom-left block shows a Jinja2 template with a 'for' loop that iterates over 'dns\_resolvers' and prints each item. Two arrows originate from the top-left block: one points to the rendered output in the top-right block, and another points to the Jinja2 template in the bottom-left block. The rendered output in the top-right block shows the 'forwarders' list with the three IP addresses, each on a new line, enclosed in curly braces and semicolons.

```
forwarders {
  1.1.1.1;
  8.8.8.8;
  9.9.9.9;
};
```

```
# template
forwarders {
  {% for n in dns_resolvers %}
  {{ n }};
  {% endfor %}
};
```

# Jinja2 conditions

```
# VM1 vars
dns_resolvers:
  - 1.1.1.1
  - 8.8.8.8
  - 9.9.9.9
use_resolvers: true
```



```
# VM2 vars
use_resolvers: false
```

```
# template
{% if use_resolvers %}
forwarders {
  {% for n in dns_resolvers %}
  {{ n }};
  {% endfor %}
};
{% endif %}
```

```
VM1:
forwarders {
  1.1.1.1;
  8.8.8.8;
  9.9.9.9;
};
```

```
VM2:
```

# Ansible facts

```
TASK [Gathering Facts] *****
ok: [my_vm]
```

```
---
- name: Play 1
  hosts: vm1
  gather_facts: no ← default is yes
  roles:
    - init
```

# Ansible variables not from /vars folder

```
---
- name: Show some variables
  hosts: vm1
  gather_facts: no # saves time, but doesn't collect info
  tasks:
    - debug:
        var: groups
    - debug:
        var: hostvars
    - debug:
        msg: This VM IP is {{ hostvars['vm1']['ansible_facts']['default_ipv4']['address'] }} # UNDEFINED

- name: Show some variables
  hosts: vm1
  tasks:
    - debug:
        msg: This VM IP is {{ hostvars['vm1']['ansible_facts']['default_ipv4']['address'] }}
```